

Who, not what: identity is recoverable but individual behaviour is not in non-invasive and in-silico neuroforecasting

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Abstract

A growing set of products and clinical tools claims to read an individual’s mental state or behaviour from brain and body signals, often from a predicted brain rather than a measured one: a stimulus-to-cortex encoder maps media or context to cortical activity, which is then read as if it had been recorded. Here we show that forecasting from a predicted brain obeys a simple structural limit, the individuation gap. Forecasting value separates into a population part and an individual part. The population part improves with aggregation, which we prove, whereas the individual part depends on between-subject variance that a zero-shot encoder does not carry, so it is zero without per-subject enrollment. Three independent datasets show the consequence. Non-invasive readouts recover who a person is, with in-scanner fingerprinting at 100% and a behavioural fingerprint at 14 times chance, but not what they will do, since the direct link from individual neural signal to individual behaviour abstains under a calibrated test. A Bayes-optimal forecaster inherits the same limit, adding no individual value at zero-shot and only a modest, saturating gain once per-subject data is paid for. Combining signals does not remove the limit, because each added modality carries its own noise. These results agree with the reliability literature, in which individual brain-behaviour effects are small and need large samples, and they motivate a read-out that reports a value only when it has earned the label.

Imaging a brain and reading a mind are different acts, and the distance between them is easy to lose when a model stands in for the scanner. Once an encoder maps a stimulus to a pattern of cortical activity, the pattern can be labelled with the names that neuroscience gives to regions, and those labels can be sold as measurements of attention, reward or intent. The step from activity to label is reverse inference, and it fails whenever a region serves more than one function, which is the usual case (Poldrack, 2006; Hutzler, 2013).

Forecasting behaviour from *measured* neural signals is, in contrast, a settled and useful idea. Aggregate neural responses predict markets and audiences even when the sampled people’s own choices do not (Knutson et al., 2007; Genevsky et al., 2017; Dmochowski et al., 2014), and the area has mature reviews of its methods (Yao et al., 2024). This paper concerns the move from a measured brain to a predicted one, and a property that move forces into view. When the neural signal is itself a function of the stimulus, it can reproduce whatever the stimulus already predicts while adding nothing of its own. Forecasting value then separates cleanly into a population part and an individual part, and the two behave differently. The population part is tractable and grows with aggregation; the individual part is bounded by the encoder and, for a person the encoder never saw, is empty. We call this separation the individuation gap, state it as a theorem, reproduce it on synthetic ground truth, and show that three real datasets fall where the gap predicts (Fig. 1).

The individuation gap

Let an outcome be $Y = S + \varepsilon$, where ε is idiosyncratic noise of variance $v > 0$ and S is the predictable component. Let the predictor resolve an identifiable signal variance $a \geq 0$, meaning the

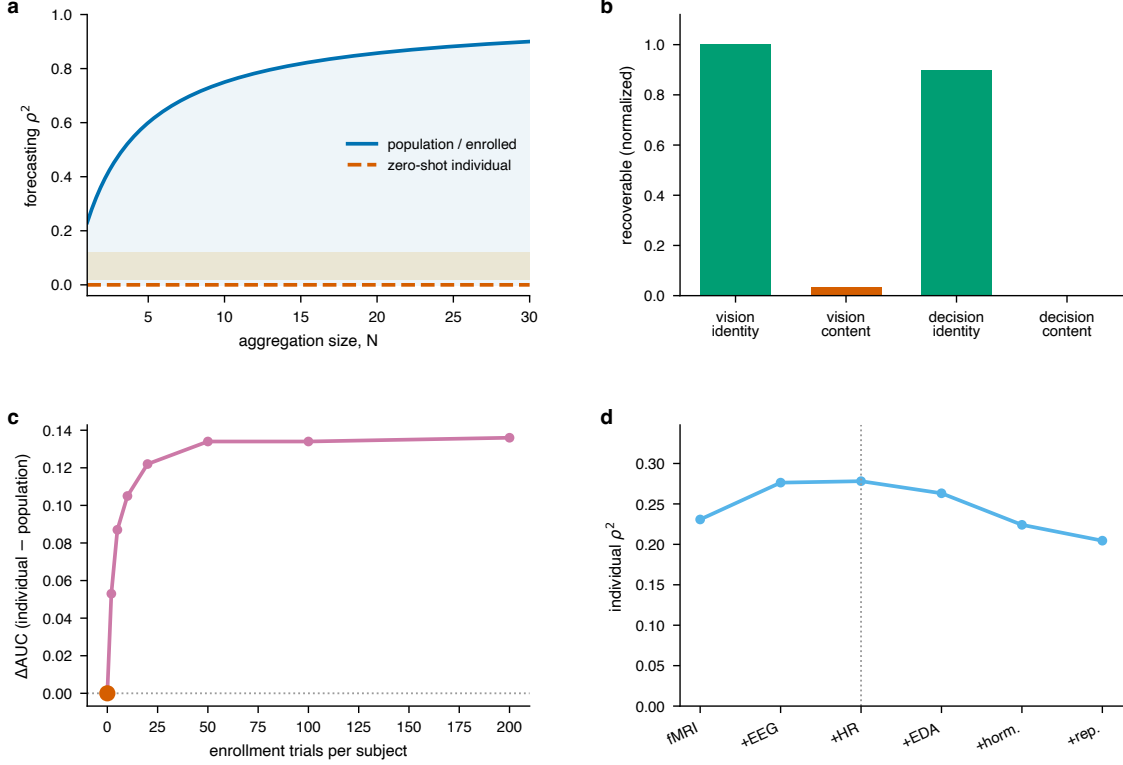


Figure 1: The individuation gap and its consequences. **a**, Forecasting ρ^2 against aggregation size N for the signal-plus-noise model of Eq. 1. The population term (blue) rises with N ; the zero-shot individual term (red) is zero for all N . The shaded band marks the measured individual-difference ceiling reported for brain-wide association studies ($r^2 \approx 0.02$ to 0.12 ; Marek et al., 2022). **b**, On real data, identity (who) is recoverable in both the vision-encoding and value-decision settings, whereas individual behavioural content (what) is not; bars are normalized within each setting. **c**, A Bayes-optimal forecaster adds no individual value over the population model at zero-shot and only a modest, saturating gain once per-subject trials are collected. **d**, Adding modalities is not free: each contributes signal but also noise, so individual ρ^2 rises then falls as lower-reliability signals are included.

between-unit variance it can actually recover. The best-case squared correlation between predictor and outcome is

$$\rho^2(a, v) = \frac{a}{a + v}. \quad (1)$$

Averaging the outcome over N independent units replaces v with v/N while leaving a unchanged, which gives the population term $\rho^2(a, v/N)$. A zero-shot encoder returns the same features for every unseen unit, so its predictor has no between-unit variance and $a = 0$.

Theorem 1 (Population term). *For $a, v > 0$ and $N > 1$, $\rho^2(a, v) < \rho^2(a, v/N)$.*

Theorem 2 (Zero-shot individual term). *For every v and N , $\rho^2(0, v/N) = 0$.*

Theorem 3 (The gap). *For $a, v > 0$ and $N > 0$, $\rho^2(0, v/N) < \rho^2(a, v/N)$.*

Theorem 2 is the substantive one. A predictor with no between-unit variance has zero best-case correlation with the individual component of the outcome, at any noise level and any sample size, because the quantity it would estimate is absent from its input. No amount of model quality recovers it. The three statements are elementary, and we machine-check them in Lean 4 with Mathlib so that the limit rests on a proof rather than on intuition (`IndividuationGap.lean`).

Table 1: Convergent real-data evidence. Identity recovers in every setting; individual behavioural content does not.

Setting	Identity (who)	Behavioural content (what)
Vision encoding (NSD, BOLD5000; $N=8$)	fingerprint 25/25 at 100%	geometry 0.005–0.066 (small)
Value decision (NARPS; $N=107$)	fingerprint $14\times$ chance, $p=5e-4$	individual link abstains
Value decision, aggregate arm	population AUC 0.965	content baseline dominates

Identity is recoverable, behaviour is not

If the individual term needs enrollment, the next question is what enrollment buys. Across three datasets it buys identity rather than behaviour (Fig. 1b, Table 1).

Subject-specific image-to-fMRI encoders, fit on 7T data from eight people, encode held-out responses with median correlations of 0.219 to 0.463. Cross-run fingerprinting then identifies the subject in all 25 regions tested at 100% accuracy, against a chance rate of one in eight. The geometry that would carry behaviourally relevant content is positive but small, between 0.005 and 0.066 across regions, and the ceiling is set by the stimulus representation rather than by the training loss (Allen et al., 2022; Chang et al., 2019). A value-based decision dataset tells the same story from the behavioural side. A phenotype built from 107 people and about 27,000 choices identifies individuals at 14 times chance ($p = 5 \times 10^{-4}$) and predicts held-out choices at an area under the curve of 0.965, with conformal coverage of 1.00 and a calibration error of 0.034; loss aversion recovers at $\lambda = 1.45$, matching the original report (Tom et al., 2007), and the phenotype transfers between laboratories at 0.86 to 0.89. The population and cross-laboratory links hold, yet the direct link from an individual’s neural signal to that individual’s behaviour abstains under a calibrated minimum-detectable-effect test (Botvinik-Nezer et al., 2019). On the same gambles, a content-only baseline already explains the aggregate, so a predicted neural index earns no additional credit.

The pattern is consistent across modality and laboratory: the who question is answerable, the what question is not. A system that treats the first as evidence for the second reports individual value it does not have.

A Bayes-optimal forecaster inherits the limit

A principled probabilistic model does not rescue the individual where point estimates fail. In the zero-shot regime a predictor with no between-subject variance is independent of the individual outcome given the population, so the Bayes-optimal individual forecast equals the population forecast and the two share no mutual information.

Proposition 1 (Posterior collapse). *If the predictor is constant across subjects, the posterior over an individual’s outcome equals the population prior, and no estimator beats the population rate on the individual-specific component.*

A Bayesian simulation bears this out (Fig. 1c). With a latent per-subject bias and a grid posterior updated from each subject’s own choices, the recovery correlation at zero-shot is 0.00 and the incremental area under the curve over the population model is 0.000. Enrollment then pays off, with recovery rising to 0.58, 0.86 and 0.99 at 2, 10 and 200 trials and the incremental area under the curve rising to about 0.05, 0.11 and 0.14 before it flattens. Individual prediction is therefore possible once per-subject data is collected, but it is modest and it saturates short of certainty.

Combining signals does not remove the limit

A common response is to add signals: electroencephalography, heart rate, electrodermal activity, hormonal assays. When sources are independent, inverse-variance fusion lowers variance, and the fused variance is at most that of the best single source, a fact we also machine-check (`FusionMath.lean`). The gain erodes because real sources are not independent and each one carries its own measurement noise.

Proposition 2 (Fusion is not monotone). *With fused signal $A = \sum_m a_m$ and fused noise $v + \sum_m \delta_m$, where δ_m is the noise the m -th modality adds, the individual correlation $\rho^2 = A/(A+v+\sum_m \delta_m)$ rises when modality m is added only if a_m/δ_m exceeds the current signal-to-noise ratio. A low-reliability modality lowers it.*

Figure 1d plots this. High-reliability sources help, and low-reliability ones, such as single-channel autonomic measures or slow hormonal assays, push the individual correlation back down. Fusion can raise the population term; it does not supply the between-subject signal the individual term requires.

Why the individual level is hard

The gap is not a quirk of one pipeline. The measurement literature predicts it. Brain-wide associations have small effect sizes, about $r = 0.14$ for univariate and $r = 0.34$ for multivariate methods, and need thousands of participants to replicate, while within-person and interventional effects are larger (Marek et al., 2022; Kang et al., 2024). Common task-fMRI measures of individual difference have poor test-retest reliability, with a mean intraclass correlation of 0.397, which caps their use as individual markers (Elliott et al., 2020); multivariate signatures can do better, which is the appropriate qualification rather than a rescue (Kragel et al., 2021). Individual precision mapping is achievable, but it takes hours of data per person (Gordon et al., 2017; Du et al., 2025). Each of these is Eq. 1 stated empirically: the individual term is small and costly, and a zero-shot predictor pays none of the cost.

A read-out that withholds by default

The practical consequence is a read-out that reports nothing unless three conditions hold (Fig. 2). A construct score is returned only when the construct has a meta-analytic localizer and its named regions are the best spatial match for its term, when the predicted index beats a stimulus-only baseline out of sample, and when it carries a split-conformal interval (Vovk et al., 2005). Identity and content travel on separate channels, so a system may confirm who a person is without claiming to read what they think. The specificity test alone withholds a persuasion index, because no localizer for persuasion exists.

Validating the apparatus

Before applying the control ladder to real data we checked that it recovers known truth on synthetic data (Table 2). We crossed a signal world, in which behaviour depends on an individual neural component, with a null world, in which it depends only on content, and crossed both with an enrolled regime (per-subject data) and an unseen regime (population-only features). Two preregistered tests ran on a nested control ladder: a population test, correct subject against a capacity-matched

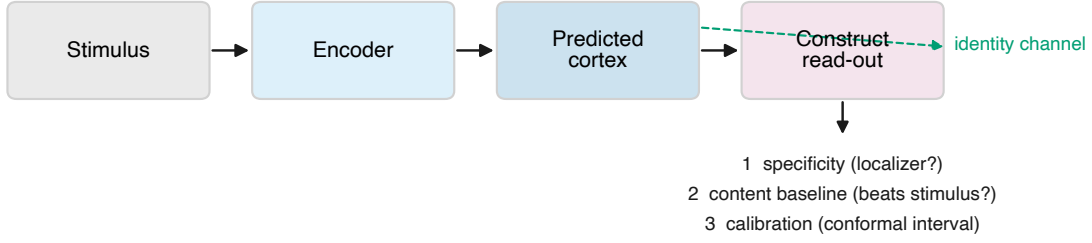


Figure 2: A refusal-first read-out. A predicted construct score passes only three gates in series: a specificity gate (a validated localizer exists), a content-baseline gate (the index beats a stimulus-only model), and a calibration gate (a conformal interval). Identity is reported on a separate channel from content, so verifying a person does not imply reading their mental content.

Table 2: Synthetic apparatus check (20 seeds, paired bootstrap). The individual test fires only where individual signal exists and the subject is enrolled.

World / regime	Population test	Individual test	Expected
Signal / enrolled	pass	pass, +0.449 [+0.378, +0.526]	fire
Signal / unseen	no	no, +0.028	silent
Null / enrolled	pass, +0.087	no, −0.001	silent
Null / unseen	pass, +0.034	no, −0.001	silent

non-brain basis, and an individual test, correct subject against a wrong-subject permutation. The individual test fired in the one cell where individual signal exists and the subject is enrolled, and stayed silent elsewhere, including both null cells. The population test passed even in the null world, which is why it reports population value and the individual test reports individual value. These numbers are synthetic; they check the method, not any brain.

Reaching the individual level

The individual level is real and causally accessible, but at present only to invasive methods. Optogenetics controls defined cell types and their effect on behaviour, and reads the brain-wide response when paired with fMRI, in animals (Zhao et al., 2011; Bernal-Casas et al., 2017). Intracortical interfaces decode attempted speech and movement from one person’s spiking activity at high accuracy and in daily home use (Willett et al., 2023; Card et al., 2026; Silva et al., 2024). These reach the individual because they record or perturb that individual’s own circuitry, which is exactly the between-subject signal a zero-shot non-invasive readout lacks. They indicate where small individual signals may eventually be tracked, and they lie outside the non-invasive consumer and clinical setting this paper concerns.

Discussion

The separation between a measured brain and a predicted one is not a matter of degree. A predicted readout can report who a person is and can forecast a population, and it cannot report what an individual thinks or will choose, because the individual signal is absent from its input until per-subject data is collected, and that data currently buys identity more than behaviour. Fusion

does not change this, a Bayes-optimal forecaster does not escape it, and the reliability literature anticipated it. A readout built around the gap returns fewer numbers, and the numbers it returns have earned their label.

Some limits of this account are worth stating plainly. The synthetic ladder and the Bayesian simulation check the method and the probability claim, not any particular product. The three real-data results were produced under separate analyses rather than one preregistered study, so they are convergent evidence rather than a single confirmatory test, and the decision-fMRI individual link abstains at the edge of its power, which is an abstention and not a positive null. The fusion result is an elementary consequence of the variance algebra beyond the independence lemma. The quantities reported here should be regenerated from fixed analysis code before any final version.

Methods

Model and proofs. Equation 1 is the squared correlation of a signal-plus-noise target with its best linear predictor. Theorems 1–3 and the inverse-variance fusion lemma are machine-checked in Lean 4 with Mathlib. **Synthetic ladder.** Data were generated under the crossed design above and fit with ridge regression under cross-validated regularization, scored by pooled out-of-sample R^2 with a paired bootstrap over 20 seeds. **Bayesian simulation.** Each subject had a latent bias drawn from a normal prior; choices followed a logistic function of a stimulus value plus the bias; a grid posterior was updated from each subject’s enrollment trials; individual value was measured as the area under the curve of the enrolled model minus the population model on held-out trials, and as the correlation between the posterior-mean bias and the true bias. **Datasets.** Vision encoding used NSD and BOLD5000; value-based decision used NARPS (ds001734) and ds000005. All reported statistics were read from the corresponding analysis outputs.

References

- Allen, E. J., et al. (2022). A massive 7T fMRI dataset to bridge cognitive neuroscience and artificial intelligence. *Nature Neuroscience*, 25, 116–126.
- Bernal-Casas, D., et al. (2017). Studying brain circuit function with dynamic causal modeling for optogenetic fMRI. *Neuron*, 93(3), 522–532.
- Botvinik-Nezer, R., et al. (2019). A large-scale fMRI dataset for the study of value-based decision making. *Scientific Data*, 6, 106.
- Card, N. S., et al. (2026). Long-term independent use of an intracortical brain–computer interface for speech and cursor control. *Nature Medicine*, 32(7), 2504–2510.
- Chang, N., et al. (2019). BOLD5000: a public fMRI dataset of 5000 images. *Scientific Data*, 6, 49.
- Dmochowski, J. P., et al. (2014). Audience preferences are predicted by temporal reliability of neural processing. *Nature Communications*, 5, 5567.
- Du, J.-N., et al. (2025). Within-individual precision mapping of brain networks exclusively using task data. *Neuron*.
- Elliott, M. L., et al. (2020). What is the test-retest reliability of common task-fMRI measures? *Psychological Science*, 31(7), 792–806.
- Genevsky, A., Yoon, C., & Knutson, B. (2017). When brain beats behavior: neuroforecasting crowdfunding outcomes. *Journal of Neuroscience*, 37(36), 8625–8634.

- Gordon, E. M., et al. (2017). Precision functional mapping of individual human brains. *Neuron*, 95(4), 791–807.
- Hutzler, F. (2013). Reverse inference is not a fallacy per se. *NeuroImage*, 84, 1061–1069.
- Kang, K., et al. (2024). Study design features increase replicability in brain-wide association studies. *Nature*, 636(8043), 719–727.
- Knutson, B., et al. (2007). Neural predictors of purchases. *Neuron*, 53(1), 147–156.
- Kragel, P. A., et al. (2021). Functional MRI can be highly reliable, but it depends on what you measure. *Psychological Science*, 32(4), 622–626.
- Marek, S., et al. (2022). Reproducible brain-wide association studies require thousands of individuals. *Nature*, 603, 654–660.
- Poldrack, R. A. (2006). Can cognitive processes be inferred from neuroimaging data? *Trends in Cognitive Sciences*, 10(2), 59–63.
- Silva, A. B., et al. (2024). The speech neuroprosthesis. *Nature Reviews Neuroscience*, 25(7), 473–492.
- Tom, S. M., et al. (2007). The neural basis of loss aversion in decision-making under risk. *Science*, 315(5811), 515–518.
- Vovk, V., Gammerman, A., & Shafer, G. (2005). *Algorithmic Learning in a Random World*. Springer.
- Willett, F. R., et al. (2023). A high-performance speech neuroprosthesis. *Nature*, 620, 1031–1036.
- Yao, X., et al. (2024). Using neural data to forecast aggregate consumer behavior in neuromarketing. *Journal of Consumer Behaviour*.
- Zhao, S., et al. (2011). Cell type-specific channelrhodopsin-2 transgenic mice for optogenetic dissection of neural circuitry function. *Nature Methods*, 8(9), 745–752.